



IMOS TCPIP Errors – Cannot Connect to AIMS

- Physical connection
- IP address
- IMOS Setup
- Services (NeopostIO / Neopost Enterprise)
- Ports Listening

AIMS Service Account

- Password Change? (must be changed by running the installer)
- Not given “Logon As A Service Rights?”
- Not given full rights to folders & files for datafile exchange, backup, autoreporting.



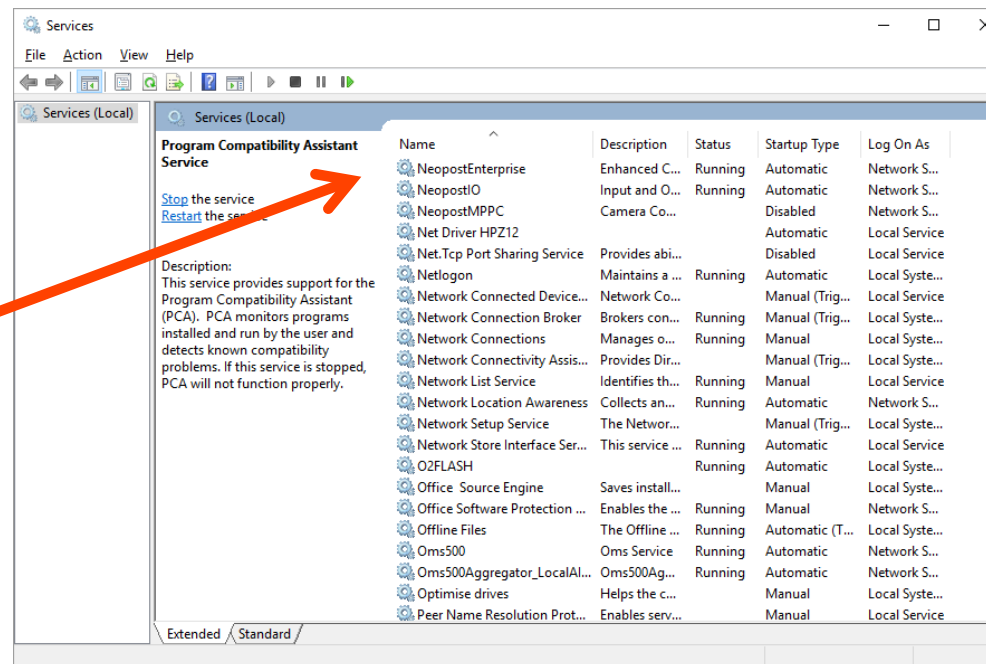
AIMS Troubleshooting: AIMS Services

AIMS Services

AIMS installs NeopostEnterprise and NeopostIO Services

Both should be running

They should run under a Service Account specified during installation. This should have access to Shared Directories and “Log on as a Service” rights





NeopostIO Files

- A new dated IO file is created each day at midnight.
- Files are located in the following AIMS directory>
 - C:\Program Files (x86)\AIMS\Services\Logs
- Notepad++ is recommended for view IO files.
- IO files DO NOT contain sensitive information.

```
Clipboard Search Navigate Bookmarks Extras Active file
NeopostIO_20210430.txt x
1 NIO»2021-04-30-09:15:38.107»JobId:»Service.Start
2 ENT»2021-04-30-09:15:38.067»JobId:»Service.Start
3 ENT»2021-04-30-09:15:40.088»JobId:»AIMS.Version:»7.4.1.1
4 ENT»2021-04-30-09:15:40.088»JobId:»Checking.License
5 ENT»2021-04-30-09:15:40.088»JobId:»Production.Connections:»3.(3.Remaining)
6 ENT»2021-04-30-09:15:40.088»JobId:»Professional.Connections:»3.(3.Remaining)
7 ENT»2021-04-30-09:15:40.088»JobId:»Office.Connections:»0.(0.Remaining)
8 ENT»2021-04-30-09:15:40.158»JobId:»Neopost.Time.Service.Server.Started
9 ENT»2021-04-30-09:15:42.368»JobId:»No.connection.could.be.made.because.the.target.machine.actively.refused.it.127.0.0.1:2003
10 ENT»2021-04-30-09:15:42.368»JobId:»HandlePreviousCrashes.--Waiting.for.NeopostIO.to.startup
11 ENT»2021-04-30-09:15:42.388»JobId:»No.connection.could.be.made.because.the.target.machine.actively.refused.it.127.0.0.1:2003
12 ENT»2021-04-30-09:15:42.388»JobId:»WorkThread.--Waiting.for.NeopostIO.to.startup
13 ENT»2021-04-30-09:15:49.420»JobId:»No.connection.could.be.made.because.the.target.machine.actively.refused.it.127.0.0.1:2003
14 ENT»2021-04-30-09:15:49.420»JobId:»WorkThread.--Waiting.for.NeopostIO.to.startup
15 ENT»2021-04-30-09:15:49.717»JobId:»No.connection.could.be.made.because.the.target.machine.actively.refused.it.127.0.0.1:2003
16 ENT»2021-04-30-09:15:49.717»JobId:»HandlePreviousCrashes.--Waiting.for.NeopostIO.to.startup
17 NIO»2021-04-30-09:15:50.029»JobId:»CommandQueue.MonitorCommandQueue.Enter
18 NIO»2021-04-30-09:15:50.045»JobId:»Checking.License
19 NIO»2021-04-30-09:15:51.185»JobId:»AIMS.Version:»7.4.1.1
20 NIO»2021-04-30-09:15:51.248»JobId:»OS.Version:»Microsoft.Windows.10.Pro
21 NIO»2021-04-30-09:15:51.248»JobId:»Production.Connections:»3
22 NIO»2021-04-30-09:15:51.248»JobId:»Professional.Connections:»3
23 NIO»2021-04-30-09:15:51.248»JobId:»Office.Connections:»0
24 NIO»2021-04-30-09:15:51.248»JobId:»Search:»True
25 NIO»2021-04-30-09:15:51.248»JobId:»Late.Diverts:»True
26 NIO»2021-04-30-09:15:51.248»JobId:»Partially.Processed.Import:»True
27 NIO»2021-04-30-09:15:51.248»JobId:»Sub.Job.Reporting:»True
28 NIO»2021-04-30-09:15:51.248»JobId:»Multi.Site.Support:»True
29 NIO»2021-04-30-09:15:51.248»JobId:»Card.Partner.Integration:»False
30 NIO»2021-04-30-09:15:51.248»JobId:»Bulk.By.Hand:»True
31 NIO»2021-04-30-09:15:51.248»JobId:»Multi.Job.Support:»True
32 NIO»2021-04-30-09:15:51.248»JobId:»Returned.Mail:»True
33 NIO»2021-04-30-09:15:51.248»JobId:»Showroom.Demo.Installation:»False
34 NIO»2021-04-30-09:15:51.248»JobId:»Enter.PerformJobMaintenance
35 NIO»2021-04-30-09:15:51.341»JobId:»CommandQueue.OnDependencyChange:»1.0
36 NIO»2021-04-30-09:15:51.341»JobId:»CommandQueue.MonitorCommandQueue.Enter
37 NIO»2021-04-30-09:15:52.888»JobId:»Exit.PerformJobMaintenance
38 NIO»2021-04-30-09:15:52.950»JobId:»<<Scanning.existing.files
```



SQL Checking Mode



SQL Checkpoint

PING

- Must use in both directions.
- PING does not use TCP or UDP or a TCP Port.
 - Ping uses ICMP protocol which may be blocked in some IT environments.

IMOS Traces and Main Event Logs

- Capturing Traces
- Main Event Logs:
 - DS200i/DS600i> C:\ProgramData\Neopost\DS244IMOS\Data\DataLogs
 - DS1200> C:\ProgramData\Neopost\DS1200 IMOS\Data\DataLogs



Netstat -a -n -o -b

- -a = displays all connections and listening ports. Such as 2002 / 2003 used for communication between IMOS and AIMS.
- -b = displays the executable involved in creating each connection or listening port. Such as NeopostIO and NeopostEnterprise services.
- -o = displays the owning process ID associated with each connection.
- -n = displays addresses and port numbers in numerical form.

Administrator: Command Prompt

```
C:\Windows\system32>netstat -a -n
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:80	0.0.0.0:0	LISTENING
TCP	0.0.0.0:135	0.0.0.0:0	LISTENING
TCP	0.0.0.0:443	0.0.0.0:0	LISTENING
TCP	0.0.0.0:445	0.0.0.0:0	LISTENING
TCP	0.0.0.0:623	0.0.0.0:0	LISTENING
TCP	0.0.0.0:690	0.0.0.0:0	LISTENING
TCP	0.0.0.0:2002	0.0.0.0:0	LISTENING
TCP	0.0.0.0:2003	0.0.0.0:0	LISTENING

Administrator: Command Prompt

```
Microsoft Windows [Version 10.0.18363.900]  
(c) 2019 Microsoft Corporation. All rights reserved.
```

```
C:\WINDOWS\system32>netstat -a -n -o -b
```

Active Connections

Proto	Local Address	Foreign Address	State	PID
TCP	0.0.0.0:80	0.0.0.0:0	LISTENING	4
Can not obtain ownership information				
TCP	0.0.0.0:135	0.0.0.0:0	LISTENING	1236
RpcSs				
[svchost.exe]				
TCP	0.0.0.0:443	0.0.0.0:0	LISTENING	4
Can not obtain ownership information				
TCP	0.0.0.0:445	0.0.0.0:0	LISTENING	4
Can not obtain ownership information				
TCP	0.0.0.0:623	0.0.0.0:0	LISTENING	14064
[LMS.exe]				
TCP	0.0.0.0:690	0.0.0.0:0	LISTENING	5648
[Oms500.exe]				
TCP	0.0.0.0:2002	0.0.0.0:0	LISTENING	5764
[NeopostEnterprise.exe]				
TCP	0.0.0.0:2003	0.0.0.0:0	LISTENING	5600
[NeopostIO.exe]				
TCP	0.0.0.0:2005	0.0.0.0:0	LISTENING	5764
[NeopostEnterprise.exe]				
TCP	0.0.0.0:2006	0.0.0.0:0	LISTENING	5600
[NeopostIO.exe]				



Group Policies

Identifying if a GP has been updated on an AIMS PC that has been added to a domain.

CMD Prompt (Admin)

```
gpresult -r
```

Group Policy is an effective way for administrators to control policy settings, deploy software, apply permissions and so on across the entire domain.

When you have multiple Group Policy Objects you need a way to verify those objects are getting applied to a user or computer.

This is exactly what GPreult was built to do.

Let's look at the example below, I have 4 group policy objects applied at different levels of the domain. One at the root, two at ADPro Computers and one at the ADPro users OU.

How do I know if they are working?

How can I check if these GPOs are getting applied correctly?

In the next section I'll show you exactly how I can use gpresult to verify these GPOs are getting applied.

```
gpresult /z
```

```
gpresult /v
```

```
gpresult /r
```



Group Policies

Parameter	Description
/s <COMPUTER>	Specifies the name or IP address of a remote computer. Do not use backslashes. The default is the local computer.
/u <USERNAME>	Uses the credentials of the specified user to run the command. The default user is the user who is logged on to the computer that issues the command.
/p [<PASSWORD>]	Specifies the password of the user account that is provided in the /u parameter. If /p is omitted, gpresult prompts for the password. /p cannot be used with /x or /h .
/user [<TARGETDOMAIN>\] <TARGETUSER>	Specifies the remote user whose RSoP data is to be displayed.
/scope {user computer}	Displays RSoP data for either the user or the computer. If /scope is omitted, gpresult displays RSoP data for both the user and the computer.
[/x /h] <FILENAME>	Saves the report in either XML (/x) or HTML (/h) format at the location and with the file name that is specified by the <i>FILENAME</i> parameter. Cannot be used with /u , /p , /r , /v , or /z .
/f	Forces gpresult to overwrite the file name that is specified in the /x or /h option.
/r	Displays RSoP summary data.
/v	Displays verbose policy information. This includes detailed settings that were applied with a precedence of 1.
/z	Displays all available information about Group Policy. This includes detailed settings that were applied with a precedence of 1 and higher.
/?	Displays Help at the command prompt.



TRACERT

How to Use the TRACERT Utility

The TRACERT diagnostic utility determines the route to a destination by sending Internet Control Message Protocol (ICMP) echo packets to the destination. In these packets, TRACERT uses varying IP Time-To-Live (TTL) values. Because each router along the path is required to decrement the packet's TTL by at least 1 before forwarding the packet, the TTL is effectively a hop counter. When the TTL on a packet reaches zero (0), the router sends an ICMP "Time Exceeded" message back to the source computer.

TRACERT sends the first echo packet with a TTL of 1 and increments the TTL by 1 on each subsequent transmission, until the destination responds or until the maximum TTL is reached. The ICMP "Time Exceeded" messages that intermediate routers send back show the route. Note however that some routers silently drop packets that have expired TTLs, and these packets are invisible to TRACERT.

TRACERT prints out an ordered list of the intermediate routers that return ICMP "Time Exceeded" messages. Using the -d option with the tracert command instructs TRACERT not to perform a DNS lookup on each IP address, so that TRACERT reports the IP address of the near-side interface of the routers.



TRACERT

How to Use the TRACERT Utility

In the following example of the tracert command and its output, the packet travels through two routers (157.54.48.1 and 11.1.0.67) to get to host 11.1.0.1. In this example, the default gateway is 157.54.48.1 and the IP address of the router on the 11.1.0.0 network is at 11.1.0.67.

The command:

```
C:\>tracert 11.1.0.1
```

The output from the command:

Tracing route to 11.1.0.1 over a maximum of 30 hops

```
-----
 1    2 ms    3 ms    2 ms    157.54.48.1
 2   75 ms   83 ms   88 ms   11.1.0.67
 3   73 ms   79 ms   93 ms   11.1.0.1
```

Trace complete.



TRACERT

How to Use TRACERT to Troubleshoot

You can use TRACERT to find out where a packet stopped on the network. In the following example, the default gateway has found that there is no valid path for the host on 22.110.0.1. Probably, either the router has a configuration problem, or the 22.110.0.0 network does not exist, reflecting a bad IP address.

The command:

```
C:\>tracert 22.110.0.1
```

The output from the command:

Tracing route to 22.110.0.1 over a maximum of 30 hops

```
-----  
1  157.54.48.1  reports: Destination net unreachable.
```

Trace complete.

TRACERT is useful for troubleshooting large networks where several paths can lead to the same point or where many intermediate components (routers or bridges) are involved.



NSLookup

NSLookup – used to resolve DNS

Quit to exit

Administrator: Command Prompt - NSLookup

```
Microsoft Windows [Version 10.0.17763.134]
(c) 2018 Microsoft Corporation. All rights reserved.

C:\WINDOWS\system32>NSLookup
Default Server:  milford-dc4.neopost.na
Address:  10.77.113.93

> Quit
Server:  milford-dc4.neopost.na
Address:  10.77.113.93

DNS request timed out.
    timeout was 2 seconds.
```

```
> help
Commands:  (identifiers are shown in uppercase, [] means optional)
NAME       - print info about the host/domain using default server
NAME1 NAME2 - as above, but use NAME2 as server
help or ?  - print info on common commands
set OPTION - set an option
    all      - print options, current server and host
    [no]debug - print debugging information
    [no]d2    - print exhaustive debugging information
    [no]defname - append domain name to each query
    [no]recurse - ask for recursive answer to query
    [no]search - use domain search list
    [no]vc     - always use a virtual circuit
    domain=NAME - set default domain name to NAME
    srchlist=N1[/N2/.../N6] - set domain to N1 and search list to N1,N2, etc.
    root=NAME  - set root server to NAME
    retry=X    - set number of retries to X
    timeout=X  - set initial time-out interval to X seconds
    type=X     - set query type (ex. A,AAAA,A+AAAA,ANY,CNAME,MX,NS,PTR,SOA,SRV)
    querytype=X - same as type
    class=X    - set query class (ex. IN (Internet), ANY)
    [no]msxfr  - use MS fast zone transfer
    ixfrver=X  - current version to use in IXFR transfer request
server NAME   - set default server to NAME, using current default server
ls server NAME - set default server to NAME, using initial server
root          - set current default server to the root
ls [opt] DOMAIN [> FILE] - list addresses in DOMAIN (optional: output to FILE)
    -a        - list canonical names and aliases
    -d        - list all records
    -t TYPE    - list records of the given RFC record type (ex. A,CNAME,MX,NS,PTR etc.)
view FILE     - sort an 'ls' output file and view it with pg
exit          - exit the program
```



Telnet

Telnet command is used to verify connection to an IP Address Port.

For example, to verify connection to 192.168.0.10 on port 25, issue the command:

telnet 192.168.0.10 25

If the connection succeeds, a blank screen will show up, meaning that the computer port is open.

A failed connection will be accompanied by an error message. It can indicate either a closed port or the fact that the indicated remote server is not listening on the provided port.



Telnet



Antimalware Service Executable – High Disk or CPU Usage



Antimalware
Service Executable



DB Utility

Local Disk (C:) > Program Files (x86) > AIMS > Utilities >

Name	Date modified	Type	Size
Scripts	3/31/2021 12:11	File folder	
AddOns.dll	3/31/2021 09:03	Application exten...	9 KB
AIMSDataStation.pfx	11/8/2019 09:56	Personal Informati...	2 KB
AIMSDataStationCert.cer	11/8/2019 09:56	Security Certificate	1 KB
DBUtility.exe	3/31/2021 09:03	Application	144 KB
firewal		Windows Batch File	3 KB
HTTPS		Application	11 KB

Open

Run as administrator



AIMS Troubleshooting: DB Utility

DB Utility – Maintenance (1)

DBUtility for AIMS v7

This DBUtility needs to be Run As Administrator
and as a user that has SQL Server SysAdmin rights (user that installed AIMS)

Maintenance (1) Standard (2) Advanced (3) File Checking (4)

Reset All Related Services

Reset All Related Services

This will reset all services including SQL Server (AIMS), NeopostIO, and NeopostEnterprise. Traffic lights should start green, then go amber, red, amber, green.

☐ ☐ ☐ **MSSQL\$AIMS**

☐ ☐ ☐ **NeopostIO**

☐ ☐ ☐ **NeopostEnterprise**

Remove Failed Auto Commands & Reset Index

Remove Commands & Reset Index

This will remove failed commands from the Command Queue where these commands were prompted automatically by Housekeeping routines. It will then reindex the table.

Stats

Generate...

Generates statistics about the command queue

Trim Command Queue

Trim Records older than 14 days

By default deletes records older than 2 weeks from the Command Queue.

Progress:

Warning - Backup AIMS before using this utility.



AIMS Troubleshooting: DB Utility

DB Utility – Standard (2)

DBUtility for AIMS v7

**This DBUtility needs to be Run As Administrator
and as a user that has SQL Server SysAdmin rights (user that installed AIMS)**

Maintenance (1) Standard (2) Advanced (3) File Checking (4)

Modify Machine Name

Current Machine Name: > New Machine Name:

This will change Machine Name in Dashboard, Analysis, and Machine List and other Drop Down menus. It will not change Machine Name in individual Job databases.

Hard Reset All Connections

This will reset all connections and check out indicators. Machines will disappear from Dashboard and will only return on next real connection. Should only be used if you are certain that all connections are not in use. If multi-site run this on all sites.

Modify Machine Operator Name

Current Operator Name: > New Operator Name:

This will change username in Alerts, Analysis and other Drop Down menus. It will not change Operator Name in individual Job databases.

Modify AIMS Username

Current Username: > New Username:

This will change username in Alerts, Analysis, Login, and Search and other Drop Down menus. It will not change usernames in individual Job databases.

Delete Test Machine

This will delete the Test machine used during factory testing and any reference to it in Analysis and selection boxes.

Delete All IMOS Log File Data

This will delete all of the data that is linked to the IMOS Log Files and the Analysis view in AIMS.

Progress:

Warning - Backup AIMS before using this utility.



AIMS Troubleshooting: DB Utility

DB Utility – Advanced (3)

DBUtility for AIMS v7

This DBUtility needs to be Run As Administrator
and as a user that has SQL Server SysAdmin rights (user that installed AIMS)

Maintenance (1) Standard (2) **Advanced (3)** File Checking (4)

Detach Old Jobs

YYYYMMDD

This will detach Jobs older than a specified date. Analysis data will be retained. Jobs with a last transaction before entered date will be detached and hidden from the Jobs list.

Delete Old AutoGen Jobs

YYYYMMDD

This will delete AutoGen Jobs older than a specified date. Analysis data will be retained. Jobs with a last transaction before entered date and starting with "AutoGen" will be deleted.

Attach All Jobs

This will attach all Jobs found in the directory. If a database has already been attached that attach will fail silently.

Delete Unused Jobs

YYYYMMDD

This will delete Jobs that have a PacksComplete of 0 and have had no transactions. It will not remove any entries that may be in Analysis.

Delete Old Jobs

YYYYMMDD

This will delete Jobs older than a specified date. Analysis data will be retained. Jobs with a last transaction before entered date will be deleted.

Remove References To Deleted Database Files

This will remove references held in SQL Server to database files that have been manually deleted.

Set Closed Jobs to SQL AUTO_CLOSE ON

This will change the AUTO_CLOSE option for each database. IVSMain, Neopost_AIMS, and all open Jobs will have AUTO_CLOSE set to OFF. All closed Jobs will have AUTO_CLOSE set to ON.

Attach Job After Permission Loss

This will attach Job that had lost permissions.

Progress:

Warning - Backup AIMS before using this utility.



AIMS Troubleshooting: DB Utility

DB Utility – File Checking (4)

DBUtility for AIMS v7

This DBUtility needs to be Run As Administrator
and as a user that has SQL Server SysAdmin rights (user that installed AIMS)

Maintenance (1) Standard (2) Advanced (3) File Checking (4)

Check NeopostIO Log

Machine: DS 1200 G3

Job Id: OJW2620676

C:\Development_Area\NeopostIO_ Check

Check NeopostIO log for inconsistencies

Progress:

Warning - Backup AIMS before using this utility.